Single-phase Boost PFC Converter - I

Introduction

In modern electrical systems, the demand for efficient and high-quality power conversion is paramount. Traditional single-phase full-bridge diode rectifiers, while simple and robust, present significant challenges when coupled with capacitive filters. As discussed in previous lectures, such configurations lead to highly distorted, peaky input currents that deviate significantly from a sinusoidal waveform. This distortion results in a non-unity power factor, meaning the system draws more apparent power than real power, leading to inefficiencies, increased losses in the grid, and potential harmonic pollution. Furthermore, to maintain a relatively constant DC output voltage, the components within these uncontrolled rectifiers, particularly diodes and capacitors, must be oversized to handle these peaky currents, adding to cost and bulk.

To address these limitations, the concept of Power Factor Correction (PFC) converters has emerged as a critical solution. These converters are designed to ensure that the current drawn from the AC source is sinusoidal and in phase with the input voltage, thereby achieving a unity power factor. This lecture introduces the single-phase boost PFC converter, a widely adopted topology that combines the rectification stage with an active power converter. By operating this power converter in a specific manner, its input impedance can be made to appear resistive, allowing for both unity power factor operation and the regulation of the output DC voltage within acceptable limits. This approach not only improves power quality but also optimizes component sizing, leading to more compact and efficient charging infrastructure.

Core Concepts

Term	Definition	Significance
`Power Factor Correction (PFC)`	A technique used to improve the power factor of an AC power system, typically by making the input current waveform more sinusoidal and in phase with the input voltage.	Essential for reducing harmonic distortion, improving energy efficiency, and complying with power quality standards in electrical systems.
	An uncontrolled rectifier circuit consisting of four	Forms the initial stage of many power converters, including the

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`Full-Bridge Diode Rectifier`	diodes arranged in a bridge configuration, converting AC input voltage into a pulsating DC output voltage.	boost PFC, by providing a rectified voltage input to the subsequent active stage.
`Boost Converter`	A DC-DC converter that steps up a DC input voltage to a higher DC output voltage, typically using an inductor, a switch, a diode, and a capacitor.	Utilized in PFC applications to regulate the output DC voltage and shape the input current, ensuring the output voltage is higher than the peak input voltage.
`Duty Ratio (D)`	The fraction of a switching period during which a power switch is turned on. It is a key control parameter in switched-mode power converters.	Allows for the control of the average voltage across the inductor and, consequently, the average current flowing through it, which is crucial for PFC.
`Switching Frequency (f_sw)`	The frequency at which the power switches (e.g., MOSFETs, IGBTs) in a converter are turned on and off.	A high switching frequency (much greater than the supply frequency) enables the assumption that the input voltage to the boost stage is nearly constant over a switching period, simplifying control.
`Supply Frequency (f_s)`	The frequency of the AC input voltage, typically 50 Hz or 60 Hz.	Represents the fundamental frequency of the AC source, against which the switching frequency is compared to establish control assumptions.
`vd`	The rectified DC voltage at the output of the full-bridge diode rectifier, which serves as the input to the boost converter stage.	This pulsating DC voltage, which is the modulus of the AC input voltage, is the reference waveform that the inductor

Term	Definition	Significance
		current is made to follow for PFC.
`V_converter (Vconv)`	The voltage at the output of the boost converter's switching stage (across the switch and diode network), which is controlled to regulate the inductor current.	By actively controlling this voltage, the boost converter can shape the inductor current to achieve unity power factor and regulate the output voltage.

Detailed Analysis

1. The Imperative for Power Factor Correction

The journey to understanding PFC begins by revisiting the limitations of a simple full-bridge diode rectifier with a capacitive filter. While this configuration provides a DC output voltage, the current drawn from the AC source is far from ideal. The presence of the capacitor causes the current to be drawn only during the peaks of the input voltage, resulting in a non-sinusoidal, peaky current waveform. This leads to a poor power factor, significant harmonic distortion, and necessitates oversized components to handle the high peak currents. The core problem is that the rectifier's input impedance is not purely resistive, which is a prerequisite for drawing a sinusoidal current in phase with the voltage.

To overcome this, a power converter is introduced between the diode rectifier and the load. The objective of this power converter is twofold: first, to present a resistive input impedance to the AC source, ensuring a sinusoidal input current in phase with the voltage (unity power factor); and second, to regulate the output DC voltage to a constant value. This active control allows for a more efficient and compliant power conversion process, making the system a "Power Factor Correction (PFC) converter."

2. Architecture of the Single-Phase Boost PFC Converter

The single-phase boost PFC converter employs a cascaded structure, combining an uncontrolled full-bridge diode rectifier with a boost DC-DC converter.
- **Full-Bridge Diode Rectifier (Front-End)**: This stage, comprising diodes D1, D2, D3, and D4, takes the AC input voltage `Vs = Vs,pk sin(ωt)` and converts it into a pulsating DC voltage, `vd`. During the positive half-cycle of `Vs`, diodes D1 and D4 conduct, making `vd = Vs`. During the negative half-cycle, diodes D2 and D3

conduct, making `vd = -Vs`. Consequently, the output voltage `vd` is always positive and represents the modulus of the input AC voltage, i.e., `vd = |Vs|`. This `vd` serves as the input to the subsequent boost converter stage.
- **Boost Converter Stage**: This stage consists of an inductor `L`, two switches `S1` and `S2` (forming a half-bridge), a capacitor `C`, and the load `RL`. The output voltage across the capacitor and load is denoted as `V_output`. The inductor `L` is placed between the rectifier output (`vd`) and the half-bridge. The key to PFC lies in actively controlling the switches `S1` and `S2` to manipulate the voltage `V_converter` at the input of the half-bridge, thereby shaping the inductor current `iL`.

3. Rectifier Output Voltage (vd) Characteristics

The voltage `vd` at the output of the full-bridge rectifier is inherently uncontrolled and directly follows the rectified AC input voltage. If `Vs = Vs,pk sin(ωt)`, then `vd = |Vs,pk sin(ωt)|`. This waveform is a series of positive half-sinusoids. The challenge for the boost converter is to draw a current `id` (which is the inductor current `iL`) that precisely follows this `vd` waveform, ensuring that the overall input current `is` (which is `id` during positive half-cycles and `-id` during negative half-cycles) remains sinusoidal and in phase with `Vs`.

4. Boost Converter Operation for PFC

The boost converter's role is to regulate the output voltage `V_output` and, crucially, to shape the input current `iL` to mimic the `vd` waveform. This is achieved by controlling the voltage `V_converter` across the half-bridge.

Key Assumption: High Switching Frequency
A fundamental assumption for the control of this converter is that the switching frequency (`f_sw`) of switches `S1` and `S2` is significantly higher than the AC supply frequency (`f_s`). This implies that a single switching period (`Ts = 1/f_sw`) is much shorter than the period of the AC input (`T = 1/f_s`).
- **Implication**: Over a single switching period `Ts`, the rectified input voltage `vd = |Vs(t)|` can be considered nearly constant. This simplification allows for easier control of the inductor current within each switching cycle.

Operating Modes within a Switching Period (Positive Half-Cycle of Vs)

Let's consider a single switching period `Ts` during the positive half-cycle of the AC input, where `D1` and `D4` are forward-biased. The duty ratio `D` represents the fraction of `Ts` for which `S1` is ON.

Mode 1: Switch S1 ON (Duration: D * Ts)
- **Circuit Configuration**: When `S1` is ON, it effectively shorts the point C to the

negative rail (ground). `S2` is OFF.
- **Inductor Voltage (vL)**: The voltage across the inductor `L` is `vL = vd - V_converter`. Since `S1` is ON, `V_converter` is approximately 0V (assuming ideal switch). Therefore, `vL = vd`. As `vd` is positive and considered constant during `Ts`, a positive voltage is applied across the inductor.
- **Inductor Current (iL)**: Since `vL = L * (diL/dt)`, the inductor current `iL` increases linearly with a positive slope: `diL/dt = vd / L`.
- **Switch Currents**: Current flows through `S1` (which is `iL`). No current flows through `S2` as it is OFF.
- **Switch Voltages**: `S2` is OFF and must block the output voltage `V_output`. The voltage across `S2` (`VS2`) will be `-V_output` (if polarity is defined as positive at the top of S2). `S1` has negligible voltage across it as it is ON.

Mode 2: Switch S2 ON (Duration: (1 - D) * Ts)
- **Circuit Configuration**: When `S2` is ON, the point C is connected to the positive output rail (`V_output`). `S1` is OFF.
- **Inductor Voltage (vL)**: The voltage across the inductor `L` is `vL = vd - V_converter`. Since `S2` is ON, `V_converter` is `V_output`. Therefore, `vL = vd - V_output`.
- **Inductor Current (iL)**: In a boost converter, the output voltage `V_output` is always greater than the input voltage `vd`. Thus, `vd - V_output` is a negative quantity. The inductor current `iL` decreases linearly with a negative slope: `diL/dt = (vd - V_output) / L`.
- **Switch Currents**: Current flows through `S2` (which is `iL`). No current flows through `S1` as it is OFF.
- **Switch Voltages**: `S1` is OFF and must block the output voltage `V_output`. The voltage across `S1` (`VS1`) will be `V_output`. `S2` has negligible voltage across it as it is ON.

Switch Realization (I-V Characteristics)
- **Switch S1**: Conducts current in one direction (positive, `iL`) when ON and blocks voltage in one direction (positive, `V_output`) when OFF. Its I-V characteristic lies in the first quadrant. This switch can be realized using a MOSFET, IGBT, or BJT.
- **Switch S2**: Conducts current in one direction (positive, `iL`) when ON and blocks voltage in the opposite direction (negative, `-V_output`) when OFF. Its I-V characteristic lies in the second quadrant. This switch can be realized using a diode or a MOSFET utilizing its body diode.

5. Achieving Unity Power Factor through Duty Ratio Control

The average inductor current `iL` over a switching period `Ts` is determined by the duty ratio `D`. By continuously varying this duty ratio `D` throughout the AC line cycle, the average value of `iL` can be made to follow the instantaneous `vd`

waveform.
- **Control Strategy**: The control system adjusts `D` such that the average `iL` in each switching cycle is proportional to `vd` at that instant. This ensures that `iL` (and thus `id`) has the same shape as `vd` (modulus of sine wave) and is in phase with it.
- **Input Current (is)**: Since `is` is equal to `id` during the positive half-cycle of `Vs` and `-id` during the negative half-cycle, and `id` is shaped to follow `|Vs|`, the resulting `is` will be a sinusoidal current in phase with `Vs`.
- **Outcome**: This precise control of the inductor current achieves unity power factor operation, where the input current is sinusoidal and perfectly aligned with the input AC voltage. The next lecture will delve into the specific mathematical relationship for how `D` must vary to achieve this.

Key Takeaways

- [ ] Uncontrolled diode rectifiers with capacitive filters draw peaky, non-sinusoidal currents, leading to poor power factor and requiring oversized components.
- [ ] Power Factor Correction (PFC) converters are used to draw sinusoidal input current in phase with the AC voltage, achieving unity power factor.
- [ ] The single-phase boost PFC converter uses a full-bridge diode rectifier cascaded with a boost DC-DC converter.
- [ ] The full-bridge rectifier converts the AC input `Vs` into a pulsating DC voltage `vd = |Vs|`.
- [ ] The boost converter actively controls the inductor current `iL` to follow the `vd` waveform, thereby shaping the input current `is`.
- [ ] A high switching frequency (`f_sw`) compared to the supply frequency (`f_s`) allows `vd` to be considered constant over a switching period.
- [ ] During the `DTs` period (S1 ON), the inductor voltage is `vd`, causing `iL` to rise.
- [ ] During the `(1-D)Ts` period (S2 ON), the inductor voltage is `vd - V_output`, causing `iL` to fall (since `V_output > vd`).
- [ ] The duty ratio `D` of the boost converter switches is dynamically varied to ensure the average `iL` tracks the `vd` waveform, leading to unity power factor.
- [ ] Switch S1 typically requires a controllable switch (MOSFET, IGBT), while S2 can be a diode or a MOSFET's body diode due to their respective I-V characteristics.

Conclusion

The single-phase boost PFC converter represents a significant advancement in power electronics, moving beyond the limitations of passive rectification to actively manage power quality. By understanding its cascaded architecture and the operational principles of its boost stage, we can appreciate how it simultaneously achieves unity power factor and regulated DC output voltage. The ability to control the inductor current by dynamically adjusting the duty ratio of the switches is central to its function, transforming a pulsating rectified voltage into a precisely shaped current that mirrors the input AC voltage. This not only enhances the efficiency and reliability of charging infrastructure but also contributes to a healthier electrical grid by mitigating harmonic distortion. The next step in this exploration will involve delving into the specific control laws and duty ratio variations required to precisely achieve these desired current waveforms, further solidifying our understanding of this crucial power conversion technology.